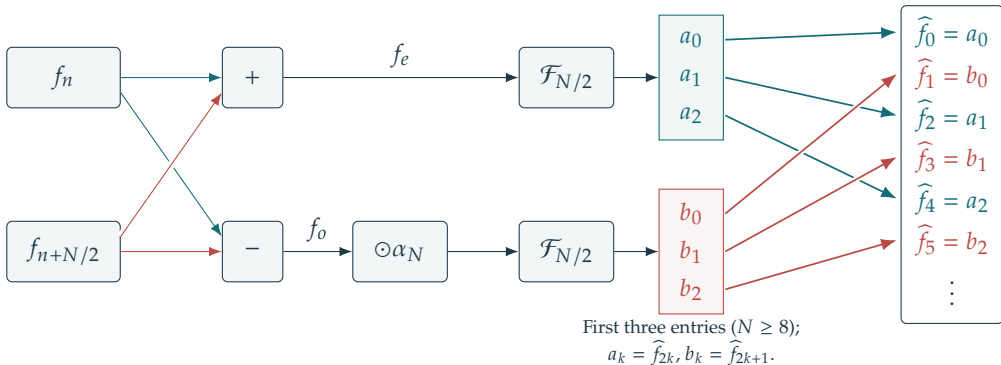


Interleaving $\mathcal{I}_N$



$$0 \leq n < N/2, \quad f_{e,n} = f_n + f_{n+N/2}, \quad f_{o,n} = f_n - f_{n+N/2}$$

$$\alpha_{N,n} = e^{-2\pi i n/N}, \quad 0 \leq k < N/2$$